

AI Integration — LLM Provider Protocol, Backends & Detection Chain

1 LLM PROVIDER PROTOCOL (Abstract Interface)

```
class LLMProvider(Protocol):  
  
    @abstractmethod  
    def score_mapping(src: str, tgt: str, context: dict) -> float  
  
    @abstractmethod  
    def classify_layer(name: str, path: str, layers: list[str]) -> str | None  
  
    @abstractmethod  
    def explain_mapping(src: str, tgt: str, signals: dict) -> str  
  
    @abstractmethod  
    def suggest_root_cause(item: str, context: dict) -> str | None  
  
    @property  
    def is_available(self) -> bool  
  
    @property  
    def provider_name(self) -> str
```

2 BACKENDS — Concrete Implementations

GitHubCopilotProvider

- REST API (copilot.github.com)
- Model: GPT-4o
- Auth: GitHub token (env var)
- Rate limit: 10 req/min
- Timeout: 30s per call
- Primary provider (weight: 0.65)

OpenAIProvider

- REST API (api.openai.com)
- Model: configurable (gpt-4-turbo)
- Auth: API key (env var)
- Rate limit: per-key tier
- Timeout: 60s per call
- Fallback provider

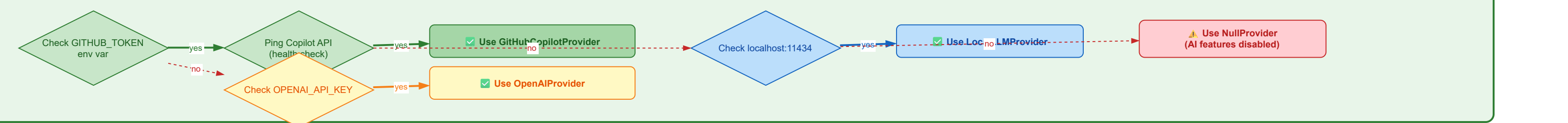
LocalLLMProvider (Ollama)

- REST API (localhost:11434)
- Model: codellama/mistral
- Auth: none (local)
- No rate limit
- Timeout: 120s (slower)
- Offline fallback

NullProvider (no AI)

- Returns 0.0 for all scores
- Returns None for classifications
- Always available
- Used when all providers fail

3 DETECTION CHAIN — Provider Discovery & Fallback



4 FIVE INTEGRATION POINTS — Where AI connects to the analysis pipeline

1 Scoring Signal Where: CompositeExternalSignalProvider Method: score_mapping(src, tgt, context) Purpose: Provide external_similarity score (weight=0.28) Fallback: Returns 0.0, coverage decreases	2 Layer Classification Where: LayerClassifier (ambiguous items) Method: classify_layer(name, path, layers) Purpose: Resolve ambiguous layer when lexicon fails Fallback: Item remains unclassified (layer=None)	3 Root Cause Tagging Where: RootCauseAnalyzer Method: suggest_root_cause(item, context) Purpose: Generate explanatory tags for anomalies Fallback: Rule-based tags only (no AI explanation)
4 AI Chat Assistant Where: AIChatPanel (GUI) Method: Free-form Q&A with analysis context Purpose: User asks questions about results Fallback: Panel hidden, feature unavailable	5 Mapping Explanation Where: ReviewPanel (on-demand) Method: explain_mapping(src, tgt, signals) Purpose: Human-readable justification for review Fallback: Shows raw signal values instead	

Design Principle: ALL AI features are gracefully degraded — the tool is 100% functional without any LLM provider configured